

XINGRONG DIAO

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RESEARCH INTERESTS

Learning-based whole-body control for humanoid robots, contact-rich interaction, deep reinforcement learning, motion planning, and sim-to-real deployment.

EDUCATION

Southern University of Science and Technology • Shenzhen, China 2023.09 – Present
M.Eng. in Electronic Science and Technology | GPA: 3.59/4.0
Thesis: Robot Navigation and Manipulation Algorithms for Unstructured Scenarios **Southern University of Science and Technology**
• Shenzhen, China 2019.09 – 2023.07
B.Eng. Robotics Engineering | GPA: 3.73/4.0

RESEARCH & PROFESSIONAL EXPERIENCE

Robotics Algorithm Intern – Airport Luggage-Transfer Robot *Ruiwei Technology* 2026.04 – 2026.06
• Developed backpack-grasping algorithms for an airport luggage-transfer robot and supported its 2026 World Robot Conference demonstration.

Humanoid Robot Learning and Whole-Body Interaction 2026.07 – Present
• Developed reinforcement- and imitation-learning policies for humanoid whole-body interaction; created **Contact-Track** to encode contact state, position, direction, object pose, and target body parts; deployed Mimic policies on hardware.

LiDAR-Based Multi-Agent Navigation *IEEE TASE* 2024.02 – 2026.08
• Designed **LSTP-Nav/LSTP-Net**, a map-free LiDAR policy combining GRU temporal modeling, attention, heading-stability reward, and physics-realistic replay; achieved **40 Hz CPU inference**.

Contact-Rich Manipulation Planning *IEEE TRO (Revise and Resubmit)* 2025.10 – Present
• Developed cascaded CSO/CPO with a surrogate contact model, mesh sampling, and prior-guided contact planning for 6D manipulation; achieved sub-ms QP solving.

GNN-Guided Motion Planning *Biomimetic Intelligence and Robotics* 2023.07 – 2024.10
• Developed a GNN-guided sampling planner with graph, obstacle, and path encoders; reduced collision checks by **99.7%** and planning time by **98.6%**.

Risk-Aware Crowd Navigation *IEEE TSMC: Systems* 2025.02 – 2026.08
• Contributed to **LR-MPC**, combining PENN risk prediction, uncertainty-aware waypoint evaluation, Multi-RRT, and MPC for socially aware crowd navigation.

SELECTED AWARDS

- **National Scholarship**, SUSTech (2025).
- **BIRob Best Paper Award** (2024).
- **MATE ROV Global Runner-Up** (2022);
- **National First Prize**, China Robot & AI Competition (2021).

SELECTED PUBLICATIONS

- [1] **Diao, X.**, Sun et al., “LSTP-Nav: Lightweight Spatiotemporal Policy for Map-Free Multi-Agent Navigation with LiDAR.” *IEEE TASE*, 2026. [\[Paper\]](#)
- [2] Sun, Z., **Diao, X.**, et al., “Socially Aware Robot Crowd Navigation via Online Uncertainty-Driven Risk Adaptation.” *IEEE TSMC: Systems*, 2026. [\[Paper\]](#)
- [3] **Diao, X.**, Chi, W., Wang, J., “Graph neural network-based method for robot path planning.” *Biomimetic Intelligence and Robotics*, 2024. [\[Paper\]](#)
- [4] Zhang, Z., **Diao, X.**, et al., “Simultaneous Contact Selection and Planning for Contact-Rich Manipulation with Cascaded Optimization.” *IEEE TRO (Revise and Resubmit)*, 2026. [\[Paper\]](#)

PATENTS & TECHNICAL SKILLS

Patents: CN219790500U (granted); CN118700132A and CN120593752A (invention).
Programming: Python, C++, MATLAB
Robotics: ROS, Gazebo, PyBullet, MuJoCo, Isaac Gym
Hardware: Franka, KUKA, Kinova **Language:** Mandarin Chinese (native), TOEFL iBT 87